

metaBRR: random-effects meta-analytic Bayesian ridge regression with an exact, `BayesianRidge`-speed fitting algorithm

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Abstract

Bayesian ridge regression (BRR) with evidence-maximised hyperparameters is a strong, tuning-free baseline for polygenic prediction. When individual-level data come from several cohorts, pooled BRR assumes a single effect vector, while per-cohort BRR wastes sample size. We describe metaBRR, which gives every cohort its own coefficient vector β_k , drawn from a learned Gaussian random-effects prior centred on a shared vector β_0 . All three variance hyperparameters are learned by type-II maximum likelihood. The key computational observation is that, for a fixed heterogeneity ratio $r = \alpha/\lambda_\delta$, integrating out the cohort deviations reduces metaBRR *exactly* to an ordinary BRR on whitened data. Hence scikit-learn’s spectral MacKay updates can be reused unchanged inside a one-dimensional search over r . In simulations with $N = 10,000$ individuals in 12 cohorts and $P = 2,000$ variants, metaBRR matches pooled BRR when effects are homogeneous. It increasingly outperforms both pooled and per-cohort BRR as cross-cohort heterogeneity grows, and it accurately recovers the cross-cohort genetic correlation.

1 Model

We observe cohorts $k = 1, \dots, K$ with genotypes $\mathbf{X}_k \in \mathbb{R}^{n_k \times P}$ and phenotypes $\mathbf{y}_k \in \mathbb{R}^{n_k}$, $N = \sum_k n_k$. Each cohort has its own intercept, handled by centring $\mathbf{X}_k$ and $\mathbf{y}_k$ within cohort. The model is

$$\mathbf{y}_k \mid \beta_k \sim \mathcal{N}(\mathbf{X}_k \beta_k, \alpha^{-1} \mathbf{I}_{n_k}), \quad (1)$$

$$\beta_k = \beta_0 + \delta_k, \quad \delta_k \sim \mathcal{N}(\mathbf{0}, \lambda_\delta^{-1} \mathbf{I}_P), \quad (2)$$

$$\beta_0 \sim \mathcal{N}(\mathbf{0}, \lambda_0^{-1} \mathbf{I}_P). \quad (3)$$

Marginally, $\text{Var}(\beta_{kj}) = \lambda_0^{-1} + \lambda_\delta^{-1}$ and $\text{Cov}(\beta_{kj}, \beta_{lj}) = \lambda_0^{-1}$ for $k \neq l$. The cross-cohort correlation of a variant’s effect, the analogue of a trans-cohort genetic correlation, is

$$\rho = \frac{\lambda_0^{-1}}{\lambda_0^{-1} + \lambda_\delta^{-1}}. \quad (4)$$

$\rho = 1$ ($\lambda_\delta \rightarrow \infty$) recovers pooled BRR. $\rho \rightarrow 0$ with $\lambda_0^{-1} + \lambda_\delta^{-1}$ fixed gives independent per-cohort BRRs that share their prior variance and noise precision. As in scikit-learn’s `BayesianRidge`, α and λ_0 get weak $\text{Gamma}(10^{-6}, 10^{-6})$ hyperpriors, and hyperparameters are set by maximising the (penalised) marginal likelihood.

2 Exact reduction to Bayesian ridge

Integrating out the deviations. Define the heterogeneity ratio $r = \alpha/\lambda_\delta = \tau_\delta^2/\sigma^2$, the per-variant heterogeneity variance in units of the noise variance. Integrating out δ_k gives

$$\mathbf{y}_k \mid \beta_0 \sim \mathcal{N}(\mathbf{X}_k \beta_0, \alpha^{-1}(\mathbf{I} + r \mathbf{X}_k \mathbf{X}_k^\top)), \quad k = 1, \dots, K \text{ independent given } \beta_0. \quad (5)$$

Let $\mathbf{W}_k = (\mathbf{I} + r \mathbf{X}_k \mathbf{X}_k^\top)^{-1/2}$, $\tilde{\mathbf{y}}_k = \mathbf{W}_k \mathbf{y}_k$ and $\tilde{\mathbf{X}}_k = \mathbf{W}_k \mathbf{X}_k$. Then $\tilde{\mathbf{y}}_k \mid \beta_0 \sim \mathcal{N}(\tilde{\mathbf{X}}_k \beta_0, \alpha^{-1} \mathbf{I})$ with $\beta_0 \sim \mathcal{N}(0, \lambda_0^{-1} \mathbf{I})$, which is *exactly* Bayesian ridge regression in (α, λ_0) on the stacked whitened data. The change of variables contributes a Jacobian, so

$$\log p(\mathbf{y} \mid \alpha, \lambda_0, r) = \log p_{\text{BRR}}(\tilde{\mathbf{y}} \mid \tilde{\mathbf{X}}; \alpha, \lambda_0) - \frac{1}{2} \sum_k \log \det(\mathbf{I} + r \mathbf{X}_k \mathbf{X}_k^\top). \quad (6)$$

Spectral form. Compute once the thin SVD $\mathbf{X}_k = \mathbf{U}_k \mathbf{S}_k \mathbf{V}_k^\top$ with $\mathbf{S}_k = \text{diag}(\mathbf{s}_k)$ of rank $q_k \leq \min(n_k, P)$. Also compute the rotated phenotypes $\mathbf{z}_k = \mathbf{U}_k^\top \mathbf{y}_k$ and the residual norm $e_k = \|\mathbf{y}_k\|^2 - \|\mathbf{z}_k\|^2$. Every whitened quantity is then diagonal in the per-cohort singular bases. Writing $d_{kj}(r) = 1/(1 + r s_{kj}^2)$:

$$\mathbf{B}(r) := \sum_k \tilde{\mathbf{X}}_k^\top \tilde{\mathbf{X}}_k = \sum_k \mathbf{V}_k \text{diag}(s_{kj}^2 d_{kj}) \mathbf{V}_k^\top, \quad (7)$$

$$\mathbf{b}(r) := \sum_k \tilde{\mathbf{X}}_k^\top \tilde{\mathbf{y}}_k = \sum_k \mathbf{V}_k (s_{kj} d_{kj} z_{kj})_j, \quad (8)$$

$$\|\tilde{\mathbf{y}}\|^2 = \sum_k (\sum_j d_{kj} z_{kj}^2 + e_k), \quad \log \det(\mathbf{I} + r \mathbf{X}_k \mathbf{X}_k^\top) = -\sum_j \log d_{kj}. \quad (9)$$

Inner loop: BayesianRidge in the eigenbasis of $\mathbf{B}(r)$. For each r we eigendecompose $\mathbf{B}(r) = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^\top$ once and set $\mathbf{c} = \mathbf{Q}^\top \mathbf{b}$. With $\mathbf{\Lambda}$ fixed, every quantity in scikit-learn's MacKay loop costs $O(P)$:

$$\hat{\mathbf{m}} = \mathbf{c} \oslash (\mathbf{\Lambda} + \lambda_0/\alpha), \quad \text{SSE} = \|\tilde{\mathbf{y}}\|^2 - 2\hat{\mathbf{m}}^\top \mathbf{c} + \hat{\mathbf{m}}^\top \mathbf{\Lambda} \hat{\mathbf{m}}, \quad \gamma = \sum_j \frac{\alpha \Lambda_j}{\lambda_0 + \alpha \Lambda_j}, \quad (10)$$

$$\lambda_0 \leftarrow \frac{\gamma + 2\lambda_1}{\|\hat{\mathbf{m}}\|^2 + 2\lambda_2}, \quad \alpha \leftarrow \frac{N - \gamma + 2\alpha_1}{\text{SSE} + 2\alpha_2}. \quad (11)$$

These updates are identical to **BayesianRidge**. At convergence the profile evidence (6) is

$$\mathcal{L}(r) = \frac{1}{2} \left[P \log \lambda_0 + N \log \alpha - \alpha \text{SSE} - \lambda_0 \|\hat{\mathbf{m}}\|^2 - \sum_j \log(\lambda_0 + \alpha \Lambda_j) - N \log 2\pi \right] + \frac{1}{2} \sum_{k,j} \log d_{kj} \quad (12)$$

plus the Gamma hyperprior terms.

Outer loop over r . $\mathcal{L}(r)$ is maximised over $\log r$. We evaluate it on a coarse log-spaced grid of 9 points spanning $[10^{-4}, 10^4]/\bar{s}^2$, where $\bar{s}^2$ is the mean squared singular value. We then refine with bounded Brent search between the neighbours of the best grid point. Finally we compare against the homogeneous limit $r = 0$, which is pooled BRR exactly. Each evaluation costs one $P \times P$ Gram accumulation and one symmetric eigendecomposition.

Cohort-specific posterior means. The joint posterior is Gaussian, and $\mathbb{E}[\delta_k \mid \beta_0, \mathbf{y}]$ is linear in β_0 . Hence $\mathbb{E}[\beta_k \mid \mathbf{y}] = \hat{\beta}_0 + \mathbb{E}[\delta_k \mid \beta_0 = \hat{\beta}_0, \mathbf{y}]$, with $\hat{\beta}_0 = \mathbf{Q} \hat{\mathbf{m}}$, and

$$\hat{\beta}_k = \hat{\beta}_0 + (\lambda_\delta \mathbf{I} + \alpha \mathbf{X}_k^\top \mathbf{X}_k)^{-1} \alpha \mathbf{X}_k^\top (\mathbf{y}_k - \mathbf{X}_k \hat{\beta}_0) = \hat{\beta}_0 + \mathbf{V}_k \left(\frac{r s_{kj}}{1 + r s_{kj}^2} (z_{kj} - s_{kj} \mathbf{v}_{kj}^\top \hat{\beta}_0) \right)_j. \quad (13)$$

Algorithm 1 metaBRR

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1: centre  $\mathbf{X}_k, \mathbf{y}_k$  within cohort; thin SVD  $\mathbf{X}_k = \mathbf{U}_k \mathbf{S}_k \mathbf{V}_k^\top$ ;  $\mathbf{z}_k \leftarrow \mathbf{U}_k^\top \mathbf{y}_k$ 
2: function PROFILE( $r$ )
3:   form  $\mathbf{B}(r)$ ,  $\mathbf{b}(r)$ ,  $\|\tilde{\mathbf{y}}\|^2$ ; eigendecompose  $\mathbf{B} = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^\top$ ;  $\mathbf{c} \leftarrow \mathbf{Q}^\top \mathbf{b}$ 
4:   run BayesianRidge MacKay updates for  $(\alpha, \lambda_0)$  in spectral form until convergence
5:   return  $\mathcal{L}(r)$ ,  $\alpha$ ,  $\lambda_0$ ,  $\hat{\beta}_0 = \mathbf{Q} \hat{\mathbf{m}}$ 
6: end function
7:  $r^* \leftarrow \arg \max_r \mathcal{L}(r)$  (log grid + Brent; compare with  $r = 0$ )
8:  $\lambda_\delta \leftarrow \alpha / r^*$ ;  $\hat{\beta}_k \leftarrow \hat{\beta}_0 + \mathbf{V}_k \text{diag}(\frac{rs}{1+rs^2})(\mathbf{z}_k - \mathbf{S}_k \mathbf{V}_k^\top \hat{\beta}_0)$ 
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Each cohort's estimate is the shared estimate plus a ridge fit of the cohort's residuals, shrunk per singular direction by $rs^2/(1+rs^2)$. Directions that are well measured in cohort k ($rs^2 \gg 1$) follow that cohort's own data; poorly measured directions fall back on $\hat{\beta}_0$. A new, unseen cohort is predicted with $\hat{\beta}_0$.

Complexity. Preprocessing is one thin SVD per cohort, $O(\sum_k n_k P q_k)$. Each outer evaluation costs $O(P^2 \sum_k q_k + P^3)$, and the inner MacKay iterations are $O(P)$ each. A naive implementation would instead work with the $(K+1)P$ -dimensional joint posterior, at $O(K^3 P^3)$ per hyperparameter update. With $N = 10^4$, $P = 2,000$ and $K = 12$, a full metaBRR fit including the ≈ 20 evidence evaluations of the r search takes a median of 22.6 s on a 12-core CPU, versus 2.5 s for scikit-learn's pooled BayesianRidge.

Verification. The unit tests (`tests/test_metabrr.py`) check three things. (i) The spectral evidence and the posterior means of β_0 and every β_k agree with a dense computation, both the explicit $N \times N$ marginal covariance and the $(K+1)P$ -dimensional joint posterior, for $n_k < P$ and $n_k > P$. (ii) The learned $(\alpha, \lambda_0, \lambda_\delta)$ are a local maximum of the dense evidence. (iii) At $r = 0$ the fit reproduces scikit-learn's BayesianRidge exactly.

3 Simulation

Design. $K = 12$ cohorts of $n_k = 833$ training individuals ($N \approx 10^4$) and 500 held-out test individuals each, with $P = 2,000$ variants. Ancestral allele frequencies are drawn from $U(0.05, 0.5)$. Cohort frequencies follow a Balding–Nichols model with $F_{ST} = 0.02$, and genotypes are standardised within cohort. Effects are $\beta_k = \sqrt{\rho} \mathbf{b}_0 + \sqrt{1-\rho} \mathbf{d}_k$ with $\mathbf{b}_0, \mathbf{d}_k \sim \mathcal{N}(0, \frac{h^2}{P} \mathbf{I})$, so every cohort has SNP-heritability h^2 and cross-cohort genetic correlation ρ . Noise variance is $1 - h^2$, and each cohort gets a random mean shift. We vary $h^2 \in \{0.1, 0.25, 0.5, 0.75\}$ and heterogeneity $1 - \rho \in \{0, 0.1, 0.25, 0.5, 0.75, 1\}$, with 3 replicates per setting.

Methods. (1) *BRR (pooled)*: BayesianRidge on the pooled, cohort-centred data. (2) *BRR (per cohort)*: a separate BayesianRidge per cohort. (3) *metaBRR*: cohort-specific $\hat{\beta}_k$. (4) *metaBRR (shared β_0)*: metaBRR's $\hat{\beta}_0$ applied to every cohort, i.e. the prediction for a new cohort. Accuracy is the within-cohort test R_g^2 , the squared correlation between predicted and true genetic value, averaged over cohorts. Phenotype R^2 is in Fig. 4.

Results. Figure 1 and Table 1 show prediction accuracy.

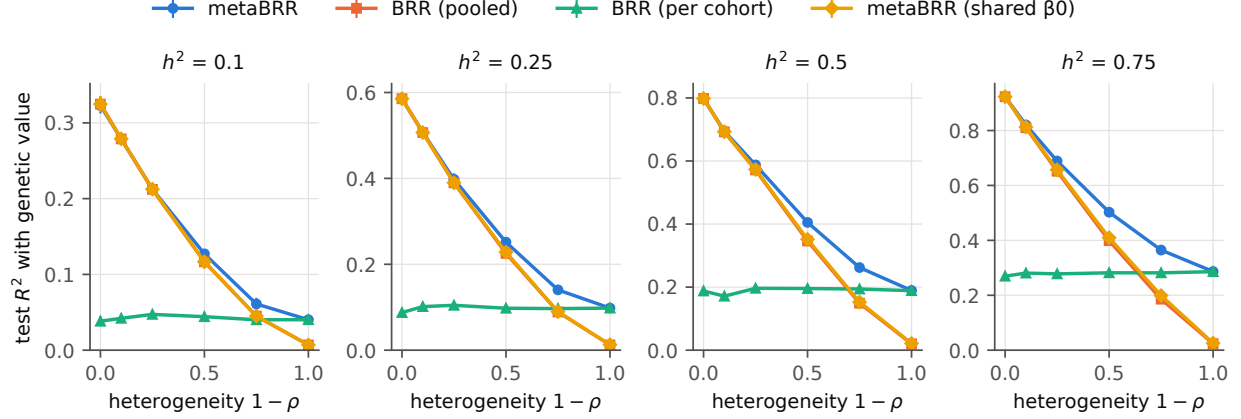


Figure 1: Test-set R^2 between predicted and true genetic value, averaged over cohorts (mean $\pm$ s.e. over 3 replicates). $N = 10^4$ in $K = 12$ cohorts, $P = 2,000$.

- *No heterogeneity* ($\rho = 1$). metaBRR learns $\hat{\rho} \approx 1$ (mean 0.96) and matches pooled BRR to within $0.006 R_g^2$ in every replicate. The extra random effect therefore costs essentially nothing when it is not needed.
- *Increasing heterogeneity*. The advantage of metaBRR over pooled BRR grows monotonically with $1 - \rho$ and with h^2 (Fig. 2). Averaged over h^2 , the R_g^2 gain is 0.015, 0.050, 0.090 and 0.138 at $\rho = 0.75, 0.5, 0.25$ and 0. At $h^2 = 0.75$ and $\rho = 0.25$, metaBRR attains $R_g^2 = 0.365$, against 0.186 for pooled and 0.282 for per-cohort BRR. At intermediate ρ metaBRR beats *both* baselines: it borrows strength across cohorts for the shared component while still fitting cohort-specific deviations.
- *No sharing* ($\rho = 0$). Pooled BRR collapses to $R_g^2 \approx 0$, while metaBRR learns $\hat{\rho} \approx 0.02$ and matches per-cohort BRR exactly.
- *Heterogeneity estimation*. The estimated cross-cohort correlation $\hat{\rho}$ tracks the truth closely across all settings (Fig. 3). It is noisier at low h^2 , where there is little genetic signal to estimate it from.
- *Shared $\hat{\beta}_0$* . Used alone, metaBRR's $\hat{\beta}_0$ predicts about as well as pooled BRR and slightly better under heterogeneity. It is the appropriate predictor for an unseen cohort.

4 Discussion and extensions

metaBRR is a drop-in generalisation of BRR for multi-cohort individual-level data. It costs one extra scalar search, and it nests pooled BRR ($r = 0$). The reduction holds for any r shared across cohorts, which requires a common noise precision α . Cohort-specific α_k break the single-ratio structure, since each cohort would then have its own $r_k = \alpha_k / \lambda_\delta$. A practical workaround is to rescale each cohort's phenotype by an estimate of its residual SD first. Natural extensions are: cohort-specific heterogeneity variances $\lambda_{\delta,k}$ (a K -dimensional outer search, or EM); structured cohort covariance, e.g. by ancestry, with $\text{Cov}(\beta_k, \beta_l) = T_{kl} \mathbf{I}$; and a summary-statistic version that replaces $\mathbf{X}_k^\top \mathbf{X}_k$ with LD matrices.

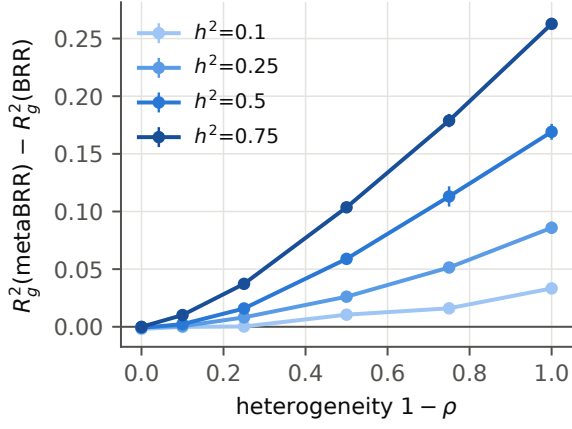


Figure 2: Gain of metaBRR over pooled BRR in R_g^2 .

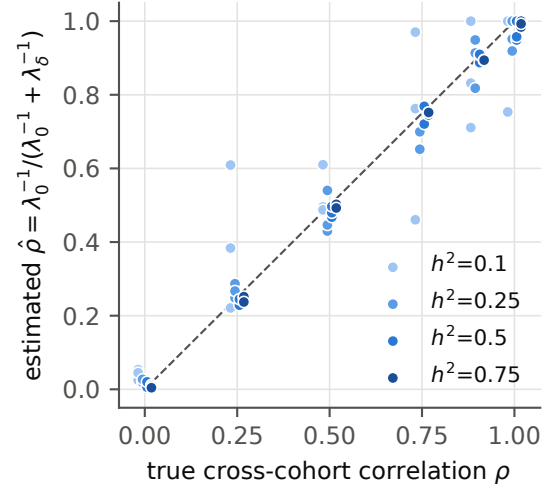


Figure 3: Estimated vs. true cross-cohort correlation (each point one replicate).

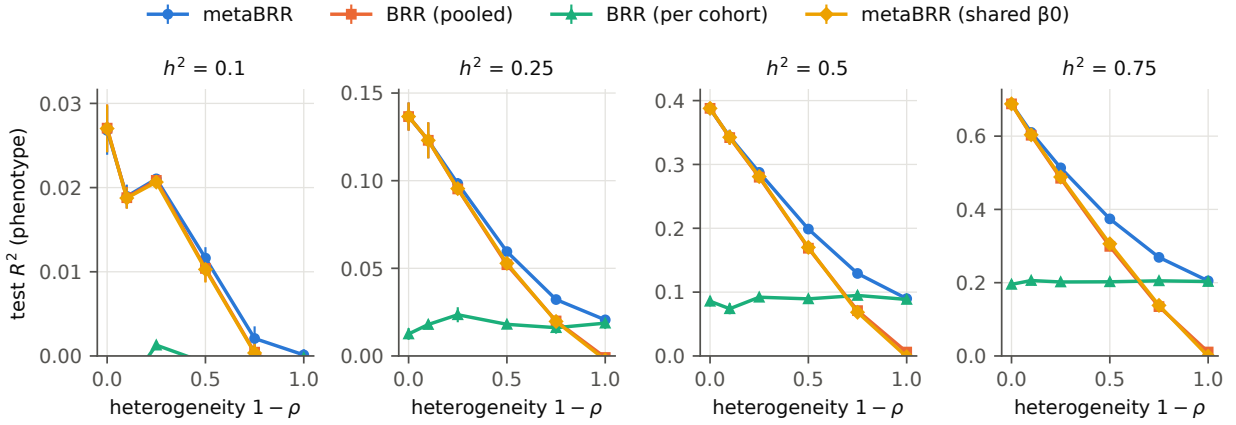


Figure 4: As Fig. 1, but test-set phenotype R^2 (upper bound h^2).

Table 1: Mean test R_g^2 (best in bold).

h^2	ρ	BRR (pooled)	BRR (per cohort)	metaBRR (shared $\hat{\beta}_0$)	metaBRR
0.1	0	0.007	0.040	0.007	0.040
0.1	0.25	0.045	0.040	0.045	0.061
0.1	0.5	0.117	0.044	0.117	0.127
0.1	0.75	0.212	0.047	0.213	0.212
0.1	0.9	0.279	0.042	0.279	0.279
0.1	1	0.325	0.038	0.325	0.323
0.25	0	0.012	0.098	0.013	0.098
0.25	0.25	0.089	0.097	0.090	0.140
0.25	0.5	0.225	0.098	0.228	0.251
0.25	0.75	0.390	0.104	0.390	0.398
0.25	0.9	0.507	0.101	0.507	0.507
0.25	1	0.585	0.088	0.585	0.585
0.5	0	0.020	0.189	0.021	0.189
0.5	0.25	0.149	0.194	0.153	0.262
0.5	0.5	0.346	0.195	0.352	0.405
0.5	0.75	0.572	0.196	0.573	0.587
0.5	0.9	0.692	0.171	0.693	0.695
0.5	1	0.798	0.188	0.798	0.797
0.75	0	0.024	0.286	0.026	0.286
0.75	0.25	0.186	0.282	0.199	0.365
0.75	0.5	0.399	0.282	0.410	0.503
0.75	0.75	0.652	0.278	0.657	0.689
0.75	0.9	0.810	0.281	0.812	0.820
0.75	1	0.923	0.270	0.923	0.923